Experiment 17: Prolog Program to Find the Sum of Integers from 1 to N

Aim:

To write a Prolog program to calculate the sum of integers from 1 to N.

Software Required:

- SWI-Prolog

Theory:

Prolog is a logic programming language used in Artificial Intelligence. The sum of integers from 1 to N can be calculated using recursion. The base case is when N equals 0, and the recursive case adds N to the sum of integers from 1 to N−1.

Algorithm:

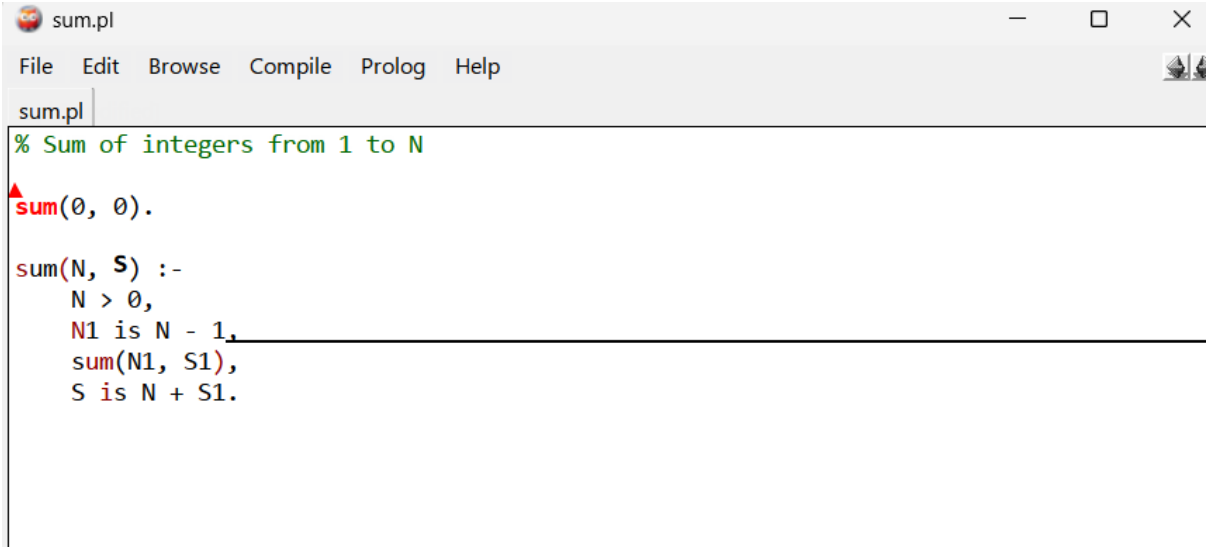
1. Start the program.
2. Read the value of N.
3. If N is 0, return 0.
4. Otherwise, calculate the sum of N and the sum of integers from 1 to N−1.
5. Display the result.
6. Stop.

Program:

```
% Sum of integers from 1 to N

sum(0, 0).

sum(N, S) :-
    N > 0,
    N1 is N - 1,
    sum(N1, S1),
    S is N + S1.
```



Output:

```
3 ?- consult('C:/Users/yadal/OneDrive/Documents/ai/1.pl').
true.


5 ?- sum(5,S).
S = 15
```

Result:

Thus, the Prolog program to calculate the sum of integers from 1 to N was implemented successfully using recursion.